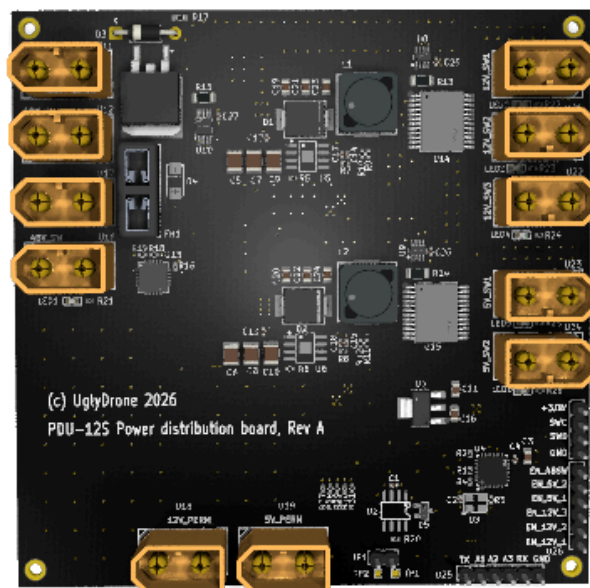


UglyDrone PDU-12S

UglyDrone PDU-12S is a smart power distribution unit for UAVs, rovers, and autonomous robotic platforms operating from a 12S / 48V battery bus.

It provides protected 12V and 5V power rails, switched high-side outputs for mission payloads, integrated voltage and current monitoring, and intelligent power sequencing.

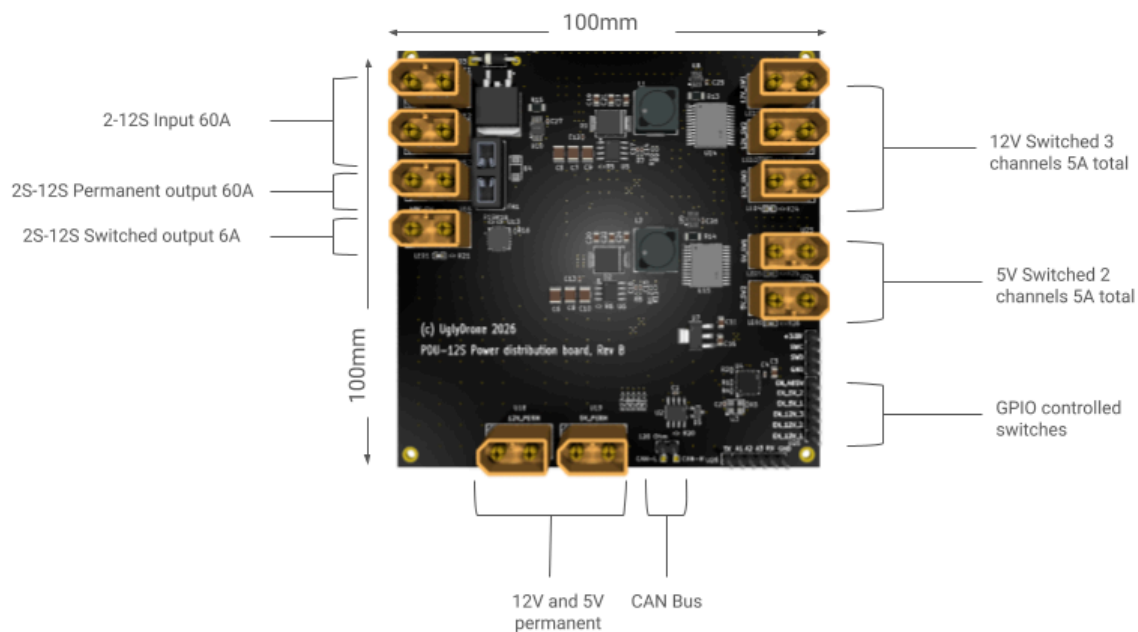
The unit enables autonomous systems to selectively power companion computers, sensors, communications, and payloads only when required, improving endurance, reliability, and serviceability.



FEATURES

- Easy to integrate into your platform, GPIO, UART and DroneCAN support
- High-side channels switching control
- Fused 48V rail
- Specially designed for airframes and robots to efficiently control power during transit segments
- ArduPilot or PX4 integration
- Voltage and current monitoring

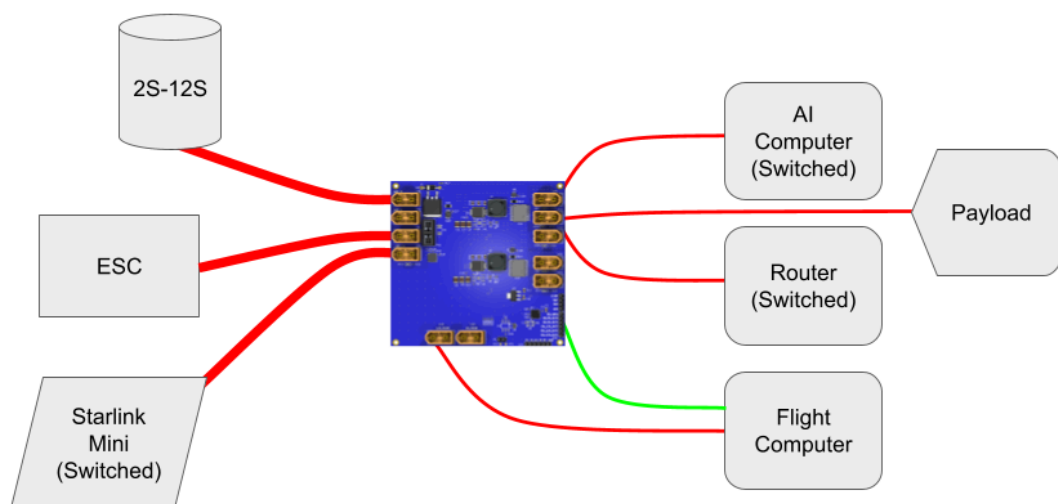
PINOUT



SPECIFICATION

Model	DroneCAN	12V current	5V current	12V switched	5V switched	48V switched
PDU-12S	No	5A	5A	3	2	1
PDU-12SPro	Yes	5A	5A	3	2	1

USAGE EXAMPLE



The **UglyDrone PDU-12S** enables mission-aware power management by integrating with an ArduPilot or PX4 flight controller through configurable GPIO outputs.

During mission planning, GPIO-controlled outputs can be assigned to power groups such as companion computers, AI accelerators, cameras, SATCOM terminals, sensors, or other payloads. At mission start, the PDU powers all required subsystems, allowing sufficient time for system initialization, GNSS acquisition, telemetry establishment, and payload self-tests. Once the vehicle reaches the long-range transit segment, non-essential equipment is automatically powered down, leaving only the flight controller and critical avionics operational to maximize endurance.

At predefined mission waypoints or in response to mission events, the autopilot can selectively restore power to communications or AI payloads for telemetry synchronization, route updates, environmental assessment, or autonomous target recognition before returning them to a low-power state. As the vehicle approaches the operational area, all required mission systems are reactivated, enabling full payload capability and, if required, establishing a low-latency operator link for supervised operation or manual mission completion.

This mission-oriented power orchestration significantly extends endurance while maintaining full autonomy and operational readiness.

CONFIGURATION CONSOLE

PDU is exposing serial port console for advanced configuration and operations. It is useful for debugging, setting advanced options and control over serial for advanced applications.

The USART console (115200 baud, 8N1) is interactive, supporting command history (up/down arrow keys) and TAB autocompletion.

- `help` - Shows the list of available commands.
- `status [N]` - Prints sensor telemetry. If `N` is specified, it refreshes automatically every `N` seconds until any key is pressed.

```
PDU> status

--- Sensor Readings ---
 5V Rail (U9 - INA219) :  5.01 V |   35 mA | 0.17 W (EN_5V_OUT1, EN_5V_OUT2)
12V Rail (U8 - INA219) : 12.00 V |   55 mA | 0.66 W (EN_12V_OUT1, EN_12V_OUT2,
EN_12V_OUT3)
48V Rail (U10 - INA228) : 48.22 V |  119 mA | 1.45 W (EN_48VSW)
```

- `outputs` - Lists all switched rails, their current state, and default startup states.
- `set [1-6] [0|1]` - Sets the state of a channel immediately (e.g. `set 1 1` turns Channel 1 ON).
- `default [1-6] [0|1]` - Configures the default startup state for a rail and saves it to Flash.
- `mode [keep|default]` - Configures startup restore behavior:
 - `keep`: Restore rails to their last active state before power loss (saves state changes dynamically to Flash).
 - `default`: Restore rails using the predefined default startup states.
- `node [1-127]` - Changes the PDU's DroneCAN Node ID (Default: 88) and saves to Flash.
- `dronecan` - Displays a live table of discovered active nodes on the bus (ID, Uptime, Health, Mode, Name, and time since last seen).

```
--- Detected DroneCAN Nodes ---
Local Node ID : 88

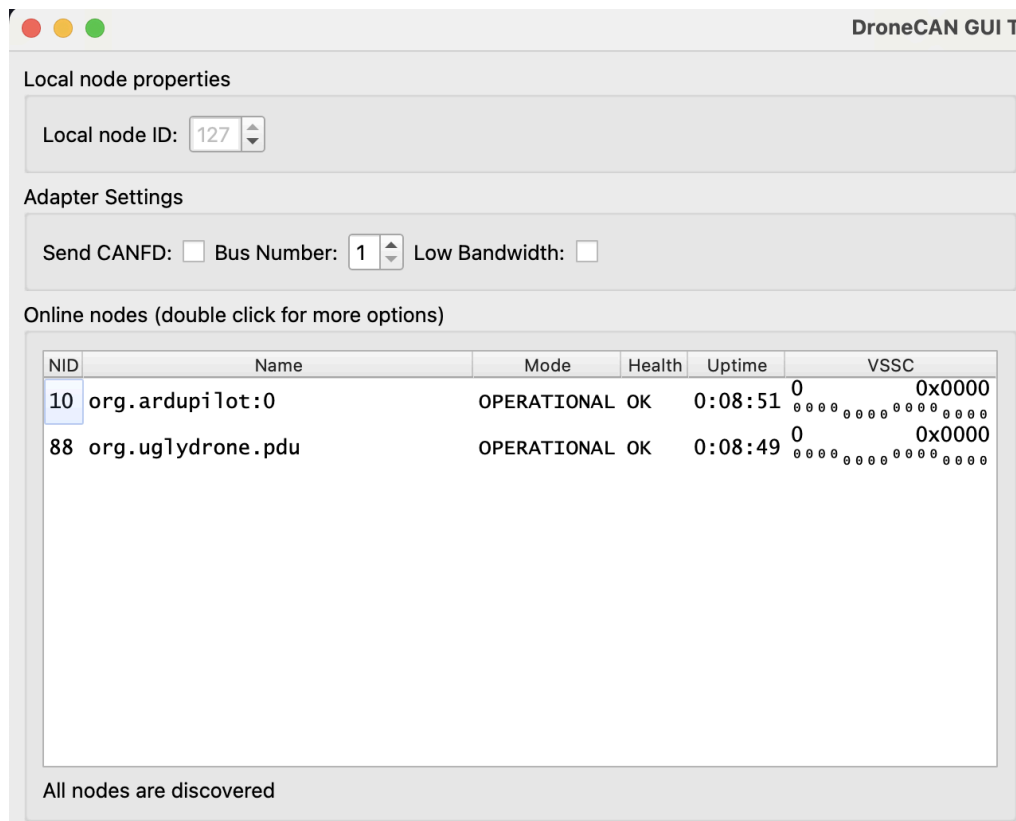
  ID | Uptime (s) | Health | Mode           | Name                               | Last Seen
-----|-----|-----|-----|-----|-----
  10 | 115        | OK     | OPERATIONAL    | org.ardupilot:0                   | 0.716 s ago
 127 | 67         | OK     | OPERATIONAL    | org.dronecan.gui_tool             | 0.057 s ago
-----|-----|-----|-----|-----|-----

PDU>
```

- `dronecan debug` - Enters command logging mode. Prints incoming hardpoint commands and service requests addressed to the PDU in real-time. Press any key to exit.
- `restart` - Reboots the PDU.

DRONECAN INTEGRATION

PDU operates on a DroneCAN bus accepting hardpoint commands to trigger outputs, once connected to a Flight Controller, check that UglyDrone PDU is presented on the bus using DroneCan GUI tool



Once visible on the bus, configure Ardupilot relays as a hardpoints

Configuration

relay

Clear

CAN_D1_UC_RLY_RT – Hz

DroneCAN relay output rate

RELAY1_DEFAULT NoChange

Relay default state

RELAY1_FUNCTION Relay

Relay function

RELAY1_INVERTED Normal

Relay invert output signal

RELAY1_PIN DroneCAN H:

Relay pin

RELAY2_FUNCTION None

Relay function

RELAY3_FUNCTION None

Relay function

RELAY4_FUNCTION None

Relay function

RELAY5_FUNCTION None

Relay function

RELAY6_FUNCTION None

Relay function

RELAY1_PIN

Cancel

Save

DroneCAN Hardpoint ID 0

Digital pin number for relay control. Some common values are given, but see the Wiki's "GPIOs" page for how to determine the pin number for a given autopilot.

Warning: Modifying values while vehicle is in flight can lead to vehicle instability and possible vehicle loss. Make sure you know what you are doing and double-check your values before Save!

☐ Advanced settings

Mapping of PDU outputs to the hardpoints

48V SW	Hardpoint 0
12V SW1	Hardpoint 1
12V SW2	Hardpoint 2
12V SW3	Hardpoint 3
5V SW1	Hardpoint 4
5V SW2	Hardpoint 5

After relays are defined, they can be used during mission planning or be mapped to RC buttons.

PDU is also sending DroneCAN telemetry in a **uavcan.equipment.power.CircuitStatus** messages. Not all Flight Controller software directly supports these messages, check manuals.